Experiment 8

Human Genetics

Outline

1. Background
2. Characteristics of DNA
3. Human Heritable Traits
4. Multiple Alleles: Human Blood Groups

Definitions

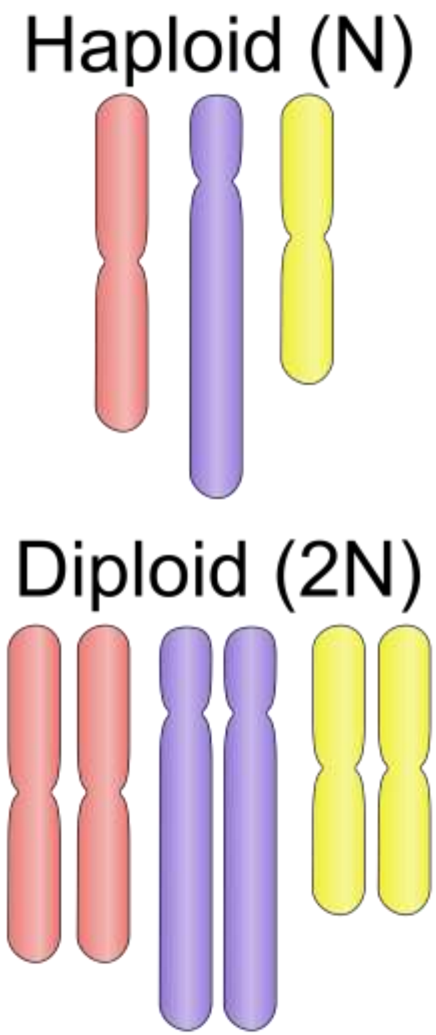
1. **Human genetics** is the study of inheritance as it occurs in human beings
2. In humans, **heredity** is the passing of heritable traits from parents to offspring
3. **Heritable traits** are passed from one generation to another via DNA

What is DNA?

- **DNA** is a long polymer that incorporates four types of bases
- The <u>sequence of bases</u> in a DNA molecule specifies the genetic information
- The **gene** is the portion of a DNA molecule that specifies a single functional unit
- The **allele** is the variant/alternative form of a gene at a <u>specific location on the chromosome</u> (*locus*)

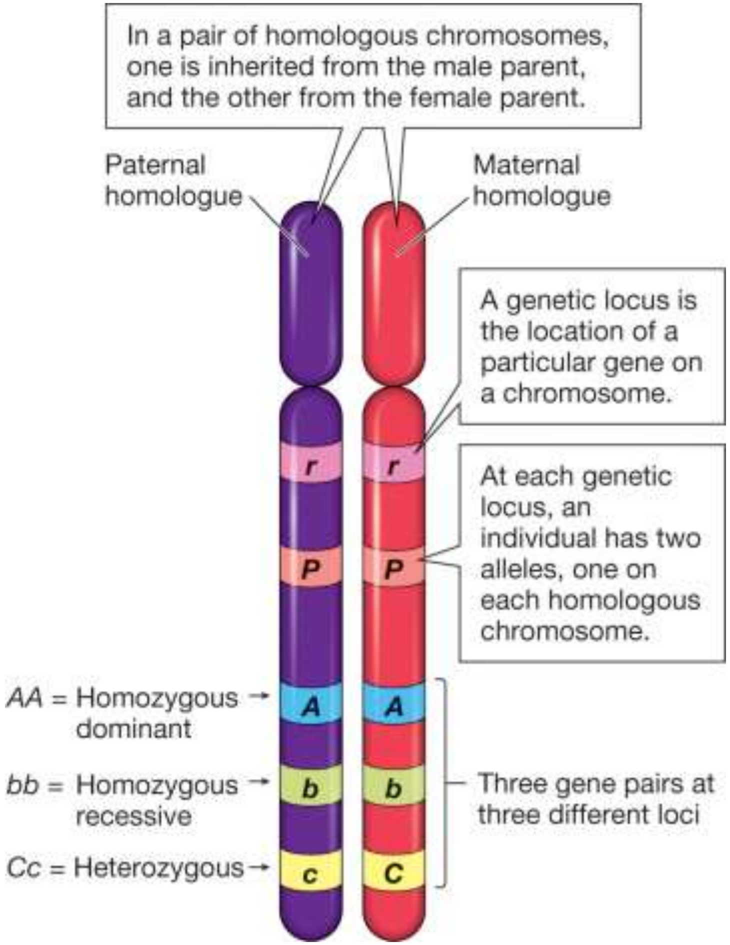
How many chromosomes do human somatic cells have?

- 46 in total (23 maternal and 23 paternal)
- **Diploid** cells have <u>double</u> the **haploid** number of chromosomes in the gamates (sex) cell
- In humans, diploid cells are <u>somatic</u> and have <u>46 chromosomes</u>, which is twice the number of chromosomes in the haploid sperm and egg cells (23)



What are homologous chromosomes?

- In diploid cells, paired chromosomes with the same gene pattern are called **homologous chromosomes**
- In each somatic cell, we have a pair of homologous chromosomes (one from the mother and one from the father)

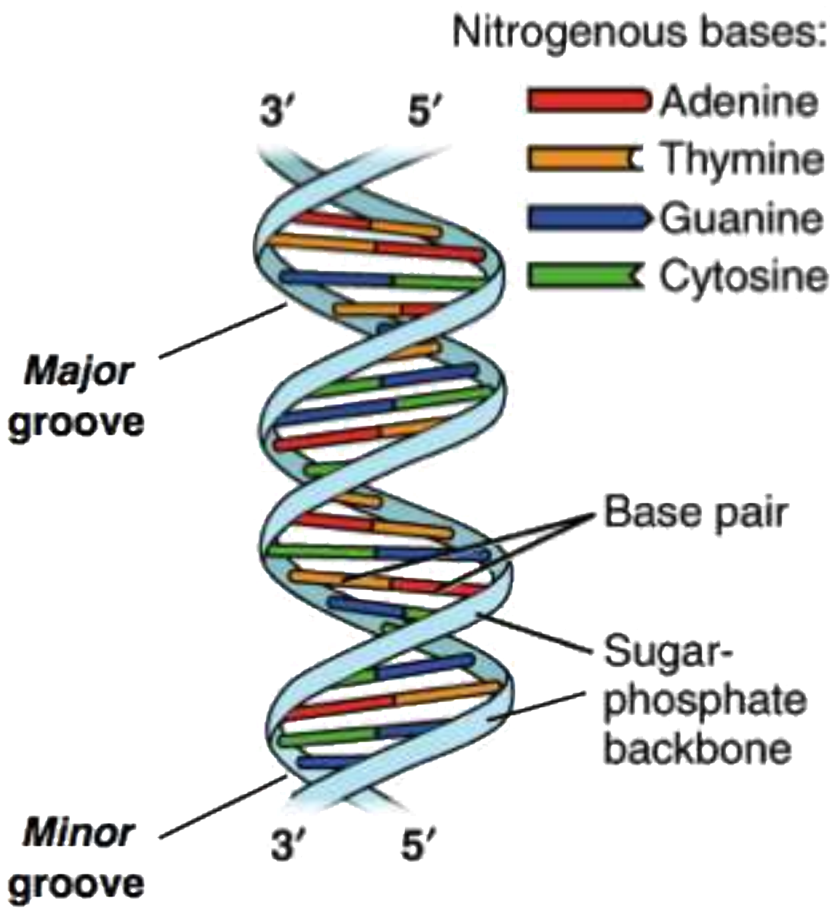


Outline

1. Background
2. **Characteristics of DNA**
3. Human Heritable Traits
4. Multiple Alleles: Human Blood Groups

Characteristics of DNA

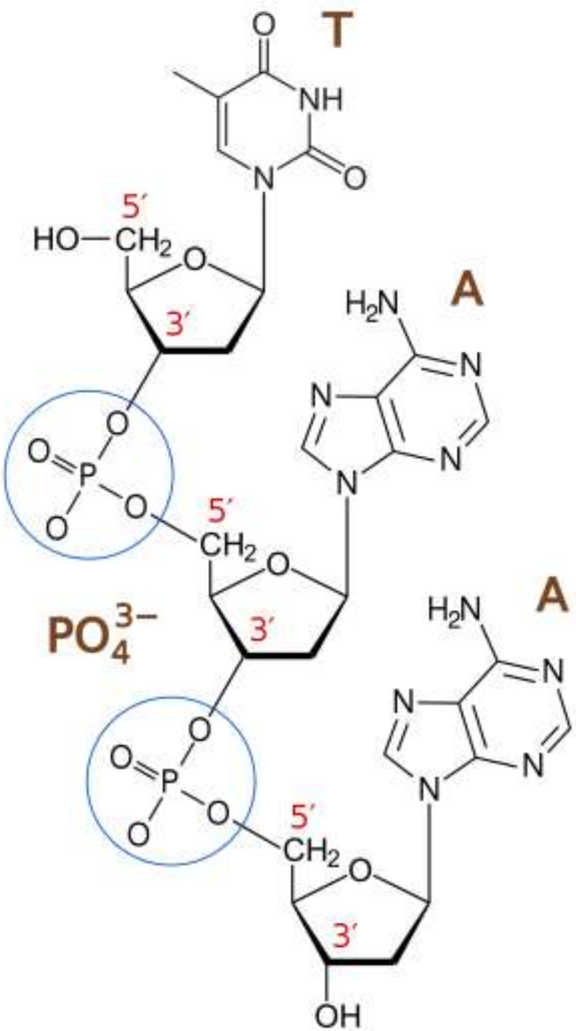
- Nucleic acid
- Consists of two long strands that are tightly held together and spiral to create a structure called the **double helix**:
  - *Outside of helix*: sugar-phosphate backbone
  - *Inside of helix*: nitrogen bases



Sugar-phosphate backbone

- **2-deoxyribose** is the sugar in DNA, and it is a *pentose* (5C) sugar

- Sugars are joined together by phosphate groups that form phosphodiester bonds between the 3rd and 5th carbon atoms of adjacent sugar rings



Nitrogen bases

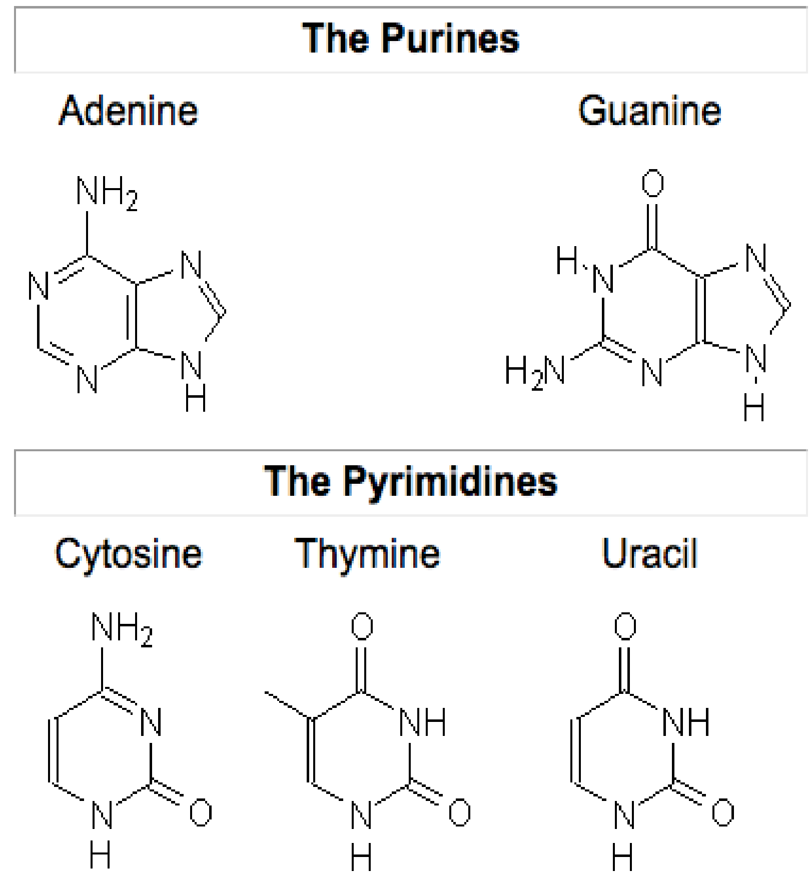
- Classified into **purines** (adenine and guanine) and **pyrimidines** (cytosine, thymine, and uracil)

- To memorize the pyrimidines and purines, learn the following mnemonics:

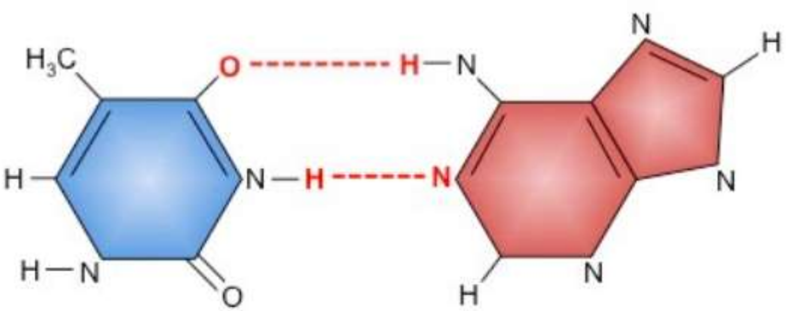
  - *For purines*: <u>Pure</u> <u>A</u>s <u>G</u>old
    - Pure for purines, A for adenine, and G for guanine
  - *For pyrimidines*: <u>Cu</u>t <u>Th</u>e <u>Py</u>e
    - C for cytosine, U for uracil, Th for thymine, and Py for pyrimidines

What is the difference between purines and pyrimidines?

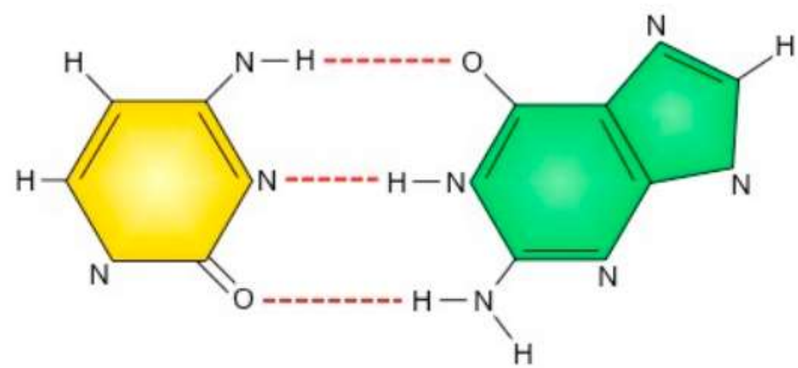
- Purines are larger than pyrimidines
- Purines are heterocyclic compounds made of fused 5- and 6-membered rings
- Pyrimidines are 6-membered rings



How do purines and pyrimidines pair?



Thymine / **U**racil pairs with **A**denine
(2 hydrogen bonds)



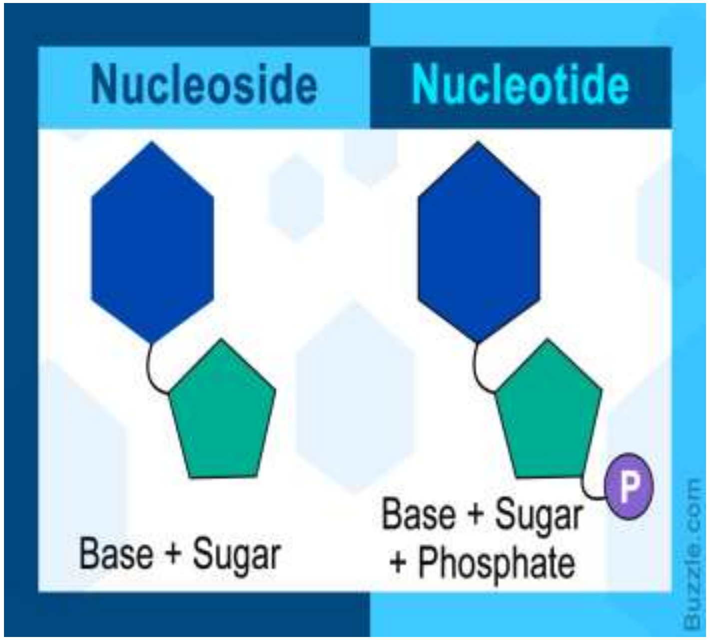
Cytosine pairs with **G**uanine
(3 hydrogen bonds)

What is the difference between DNA and RNA?

DNA	RNA
Deoxyribonucleic acid	Ribonucleic acid
Sugar is 2-deoxyribose (*pentose*)	Sugar is ribose (*pentose*)
Nitrogen bases are A, G, C, and T	Nitrogen bases are A, G, C, and U
Double-stranded	Single-stranded

What is the difference between a nucleoside and a nucleotide?

- **Nucleoside =** nitrogen base + 5C sugar
- **Nucleotide =** nitrogen base + 5C sugar + phosphate group
- The monomer of DNA is the **nucleotide**



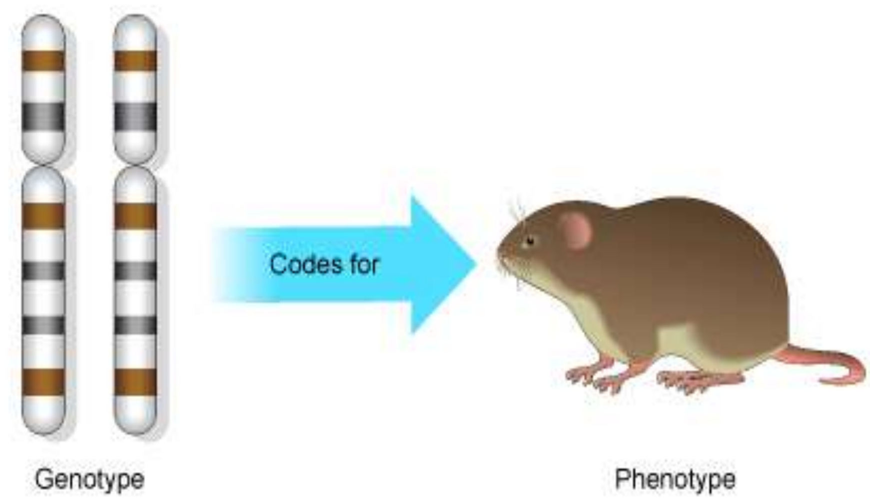
Outline

1. Background
2. Characteristics of DNA
3. **Human Heritable Traits**
4. Multiple Alleles: Human Blood Groups

Phenotype vs Genotype

- The **phenotype** of an organism is the complete set of observable traits of its structure and behavior

- The **genotype** of an organism is the complete set of genes within its genome



What is the relationship between genes and traits?

1. **Single-gene traits**
   1. ***One gene controls one trait***
      i. **Two alleles only**
      ii. **More than two alleles are involved (multiallelic)**
   2. ***One gene controls more than one trait (pleiotropy)***
2. **Multigenic traits**
   1. ***Epistasis***
      - **One gene at a certain locus controls the expression of another gene at another locus**
   2. ***Polygeny***
      - **More than one gene controls the trait, and the trait is quantitative, meaning that the phenotypes vary in degree**

Human skin color is an example of a polygenic trait

Gene 1	dd	dD	dD	DD	Dd	Dd	DD
Gene 2	dd	dd	dD	Dd	Dd	DD	DD
Gene 3	dd	dd	dd	dd	DD	DD	DD
	 Very light			 Medium			 Very dark
# of light "d" alleles	6	5	4	3	2	1	0
# of dark "D" alleles	**0**	**1**	**2**	**3**	**4**	**5**	**6**

What is the relationship between alleles and traits?

1. **Complete dominance**
   - Dominant gene expresses itself in homozygous dominant (RR) or heterozygous (Rr) genotypes
   - Recessive gene expresses itself only in homozygous recessive (rr) genotypes
   - *Ex*: RR = red; rr = white; Rr = red
2. **Incomplete dominance**
   - Both alleles express themselves partially to give an intermediate
   - *Ex*: RR = red; rr = white; Rr = pink
3. **Codominance**
   - Both alleles express themselves fully
   - *Ex*: $I^A I^B$ = AB blood group

How are traits inherited?

1. **Autosomal inheritance**
   - Genes are carried on the 22 autosomal chromosomes
   - Genes on different chromosomes are inherited independently via *independent assortment*
   - Genes on the same chromosome are *linked*, meaning that they are inherited together <u>unless</u> crossing over occurs
2. **Sex-linked inheritance**
   - Genes carried on the sex chromosomes (X or Y)
3. **Sex-affected inheritance**
   - Genes carried on the 22 autosomal chromosomes, but their expression is affected by sex

The Practical part

Single-gene traits

Autosomal traits				
Trait and Description	**Dominant**	**Genotype**	**Recessive**	**Genotype**
Tongue roller Ability to curl up the side of their tongue to form a tube shape.		TT , Tt		tt
Widow's peak If your hairline forms a point at the center of the forehead, you have a widows peak. If not, you a straight hairline.		WW, Ww		ww
Attached earlobe If earlobes hang free, they are detached. If they connect directly to the sides of the head, they are attached.		AA, Aa		aa

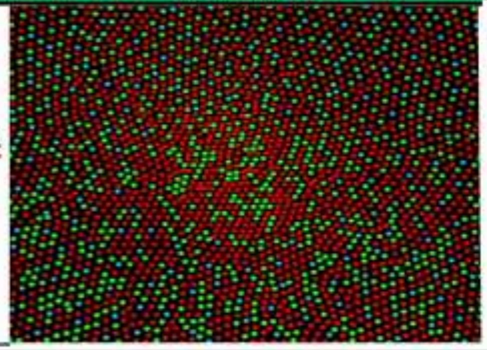
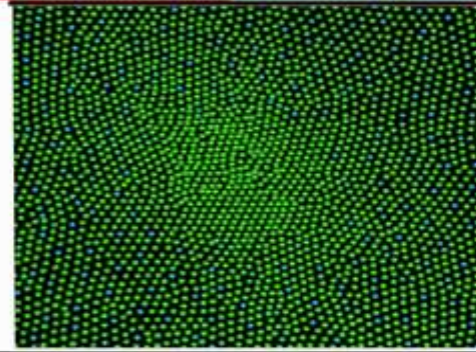
Single-gene traits

Autosomal traits				
Trait and Description	**Dominant**	**Genotype**	**Recessive**	**Genotype**
Freckles Freckles are small, concentrated spots of a skin pigment called melanin. Freckles show a domin -ant inheritance pattren.		FF, Ff		ff
Has dimples Dimples are small, natural indentations on the cheeks. they can appear on one or both sides.		HH, Hh		hh
Mid-digital hair Hair growing in the mid- digit of each finger is dominant over no hair in the mid-digit.		MM, Mm		mm

Single-gene traits

Autosomal traits				
Trait and Description	**Dominant**	**Genotype**	**Recessive**	**Genotype**
Cleft chin Small natural indentations on the chin, control by dominant gene. No cleft chin is recessive.		LL, Ll		ll
Hitchhiker's thumb Is the ability to bent the thumb back-ward. It is control by recessive single gene. Unable to do that is dominant.		II, Ii		ii
Bent Pinky Bent little finger is dominant, while Straight little finger is recessive.		EE, Ee		ee

X- Linked traits

Trait and Description	Dominant	Genotype	Recessive	Genotype
Color blindness Unable to see red, green, or blue light and this means they mix up all color which have some red or green as a part of the whole color. It is caused by a single recessive gene located on the X - chromosome.		Male: $X^c Y$ Female: $X^c X^c$		Male: $X^C Y$ Female: $X^C X^C$ $X^C X^c$ Carrier

Sex affected traits

Trait and Description	Dominant	Genotype	Recessive	Genotype
Baldness (Hair loss) Affected by sex, the gene for this trait is dominant in male, needs one gene to appear (appear in Homozygous and Heterozygous). In female, the gene is recessive,the trait needs two gene to appear (only in Homozygous state).		Male: ZZ, HZ Female: ZZ		Male: HH Female: HH, HZ
Index finger is shorter than ring finger The index (2nd finger counting the thumb as number one) finger is shorter than the ring (4th) finger. The expression of this gene is sex influenced. The phenotype is usually dominant in males and recessive in females.		Male: II, IR Female: II		Male: RR Female: RR,IR

What are X-linked disorders?

- Caused by a single recessive gene located on the X chromosome
- Phenotype is expressed in males with the recessive gene and in females who are homozygous for this recessive gene
- Examples of X-linked disorders include color blindness and hemophilia

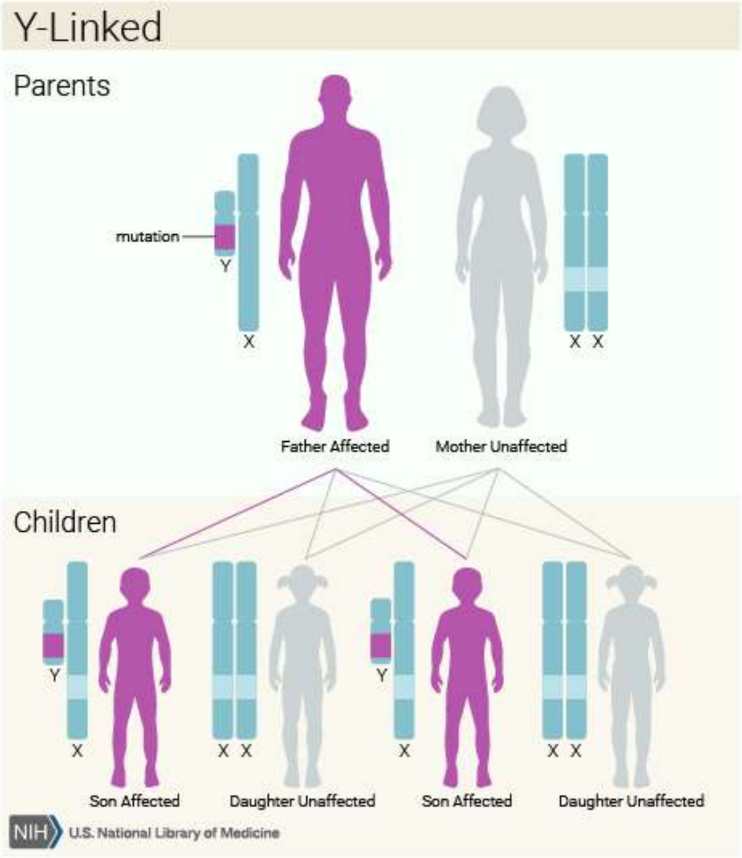
Why are they rarely seen in females?

- Females have two copies of the X chromosome:
  - X^BX^B = normal vision
  - X^BX^b = normal vision
  - X^bX^b = colorblind
- Males have only one copy of the X chromosome:
  - X^BY = normal vision
  - X^bY = colorblind

When does Y-linked inheritance occur?

- When a gene, trait, or disorder is transferred through the Y chromosome

- Y-linked traits are only passed from father to son



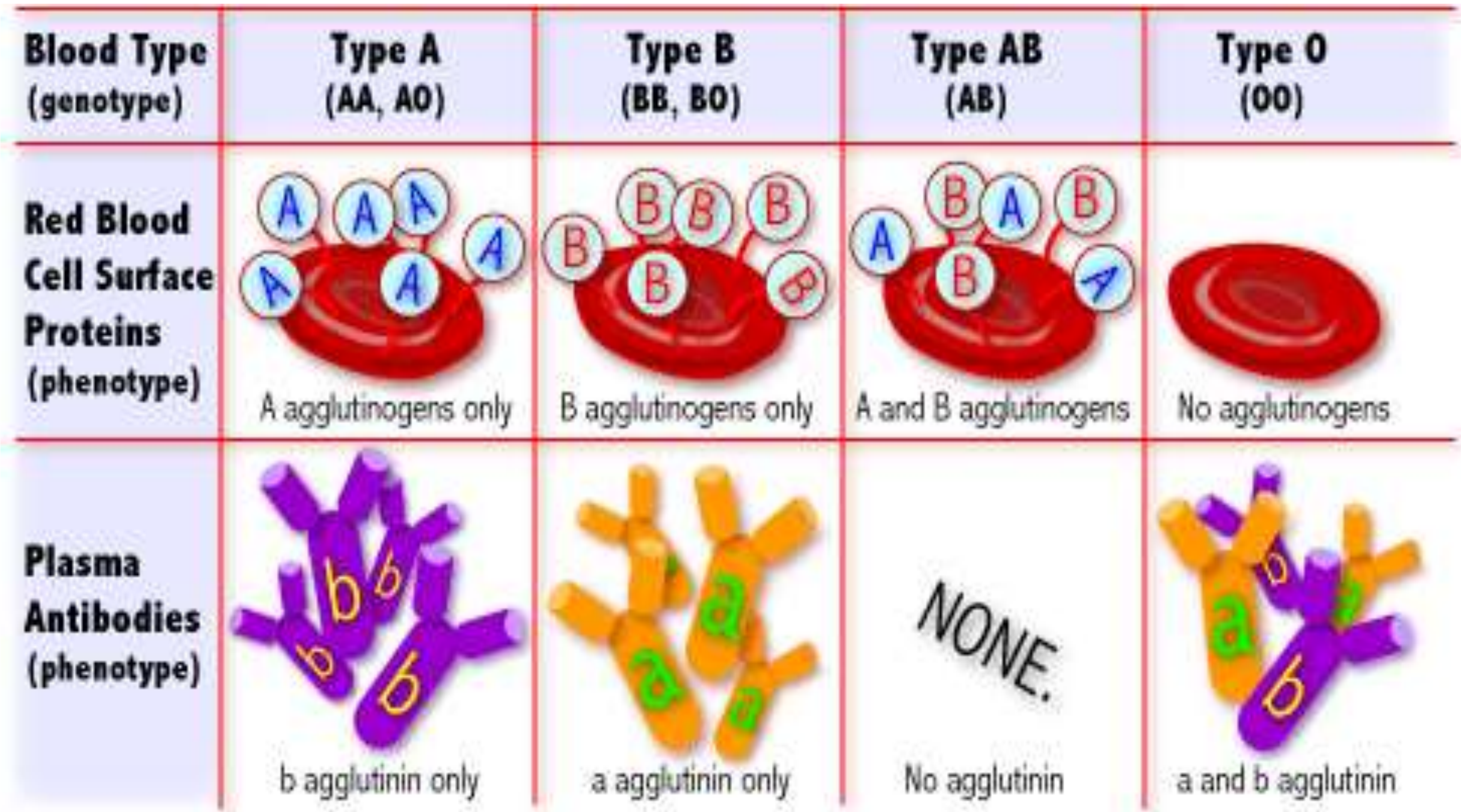
Outline

1. Background
1. Characteristics of DNA
2. Human Heritable Traits
3. **Multiple Alleles: Human Blood Groups**

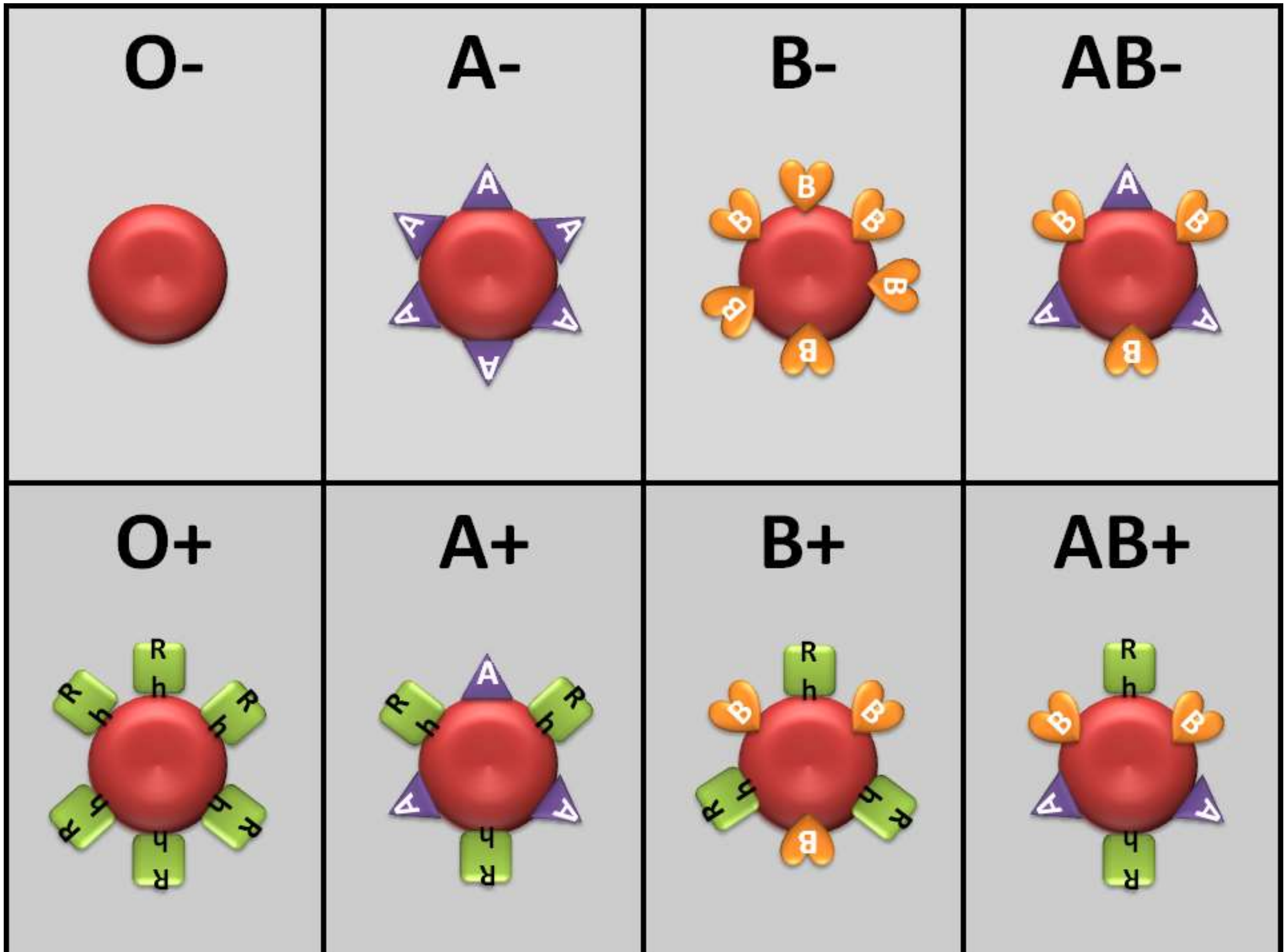
Blood Type

- **Blood type** is the classification of blood based on the presence or absence of inherited antigenic substances on the surface of RBCs
- These inherited antigenic substances can be proteins, carbohydrates, glycoproteins, or glycolipids depending on the blood group system
- The two most important blood groups in humans are:
  1. **ABO** = A, B, AB, or O
  2. **RhD** = +ve or -ve

ABO Blood Group System



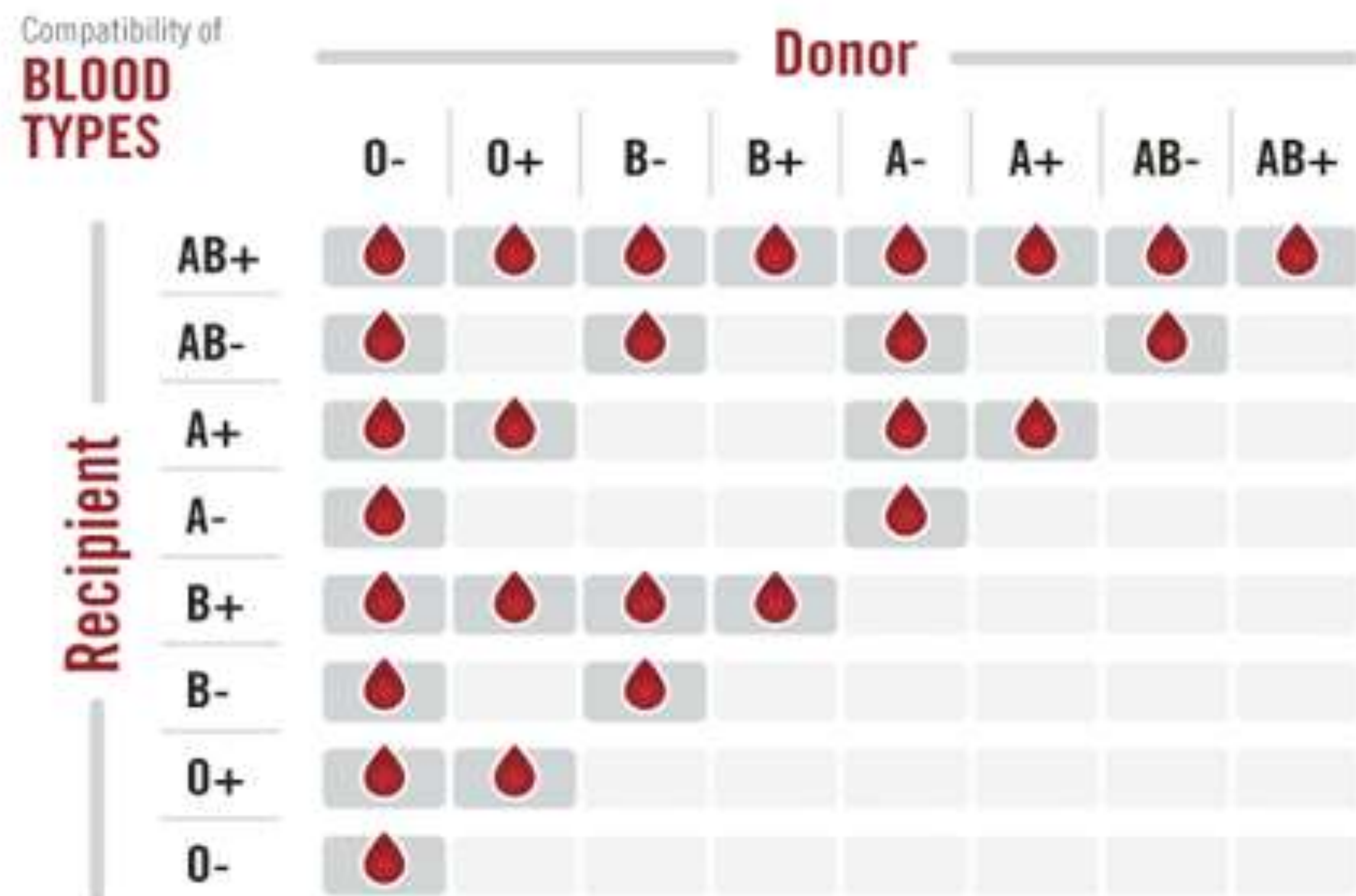
Rh D Blood Group system



ABO and Rh D Genotypes

Phenotype	Genotype	Antigens on RBC	Antibodies in plasma
A	I^AI^A, I^Ai	A	anti-B
B	I^BI^B, I^Bi	B	anti-A
AB	I^AI^B	AB	None
O	ii	None	anti-A and anti-B
Rh^+	Rh^+Rh^+	Rh	None
Rh^+	Rh^+Rh^-	Rh	None
Rh^-	Rh^-Rh^-	None	anti-Rh

Blood Donation



Universal Donor and Recipient

- *Universal donors* are those with the <u>blood group O,</u> because they can donate blood to individuals of any ABO blood group. However, they can only receive blood from group O individuals.
- *Universal recipients* are those with <u>blood group AB</u>, because they can receive blood from individuals of any ABO blood group. However, they can only donate blood to group AB individuals

Blood Typing

- **Blood typing:**

is used to determine an individual's blood group (to establish whether a person is blood group A, B, AB, or O and whether he or she is Rh^+ or Rh^-)

Determining ABO Blood Type

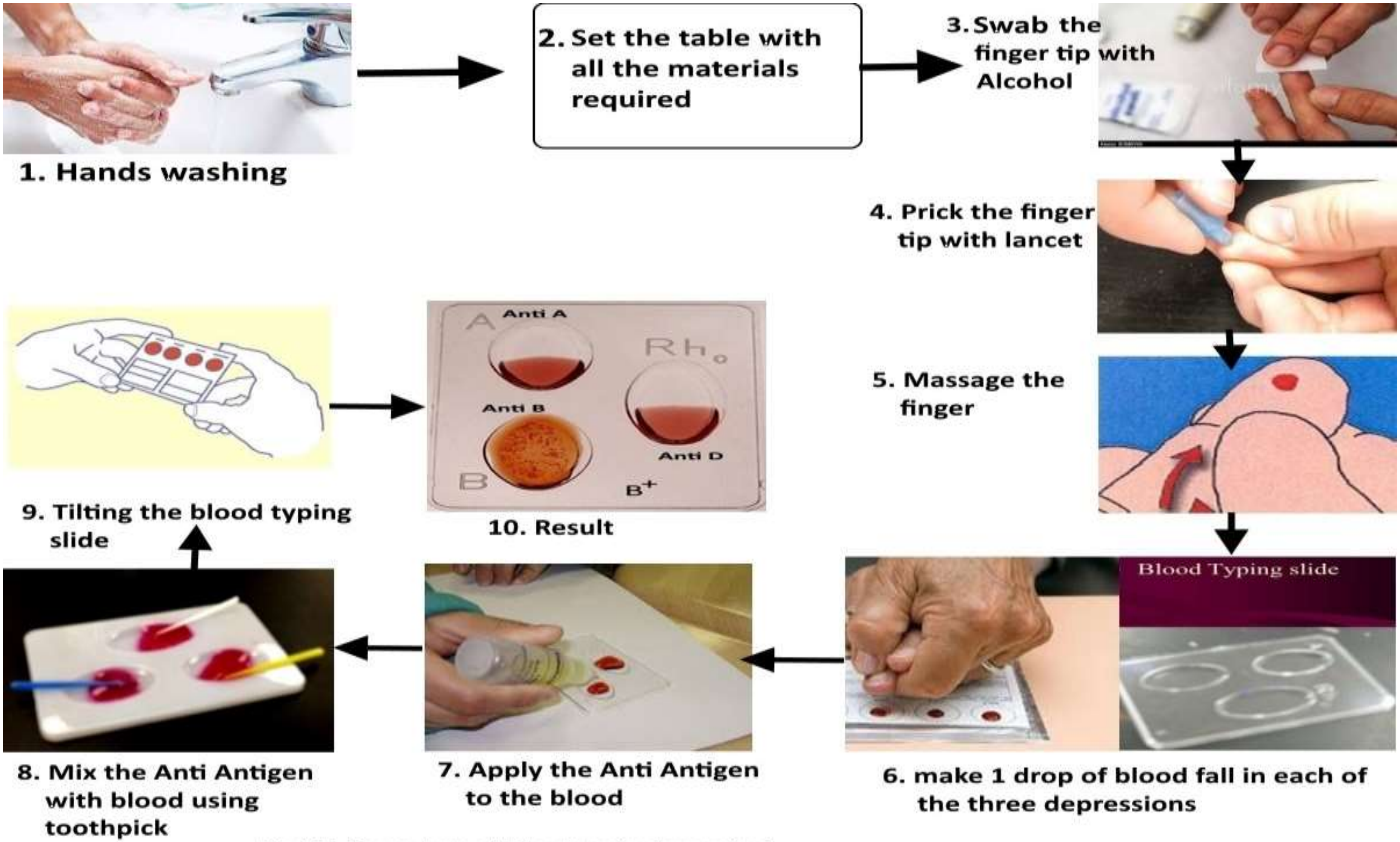
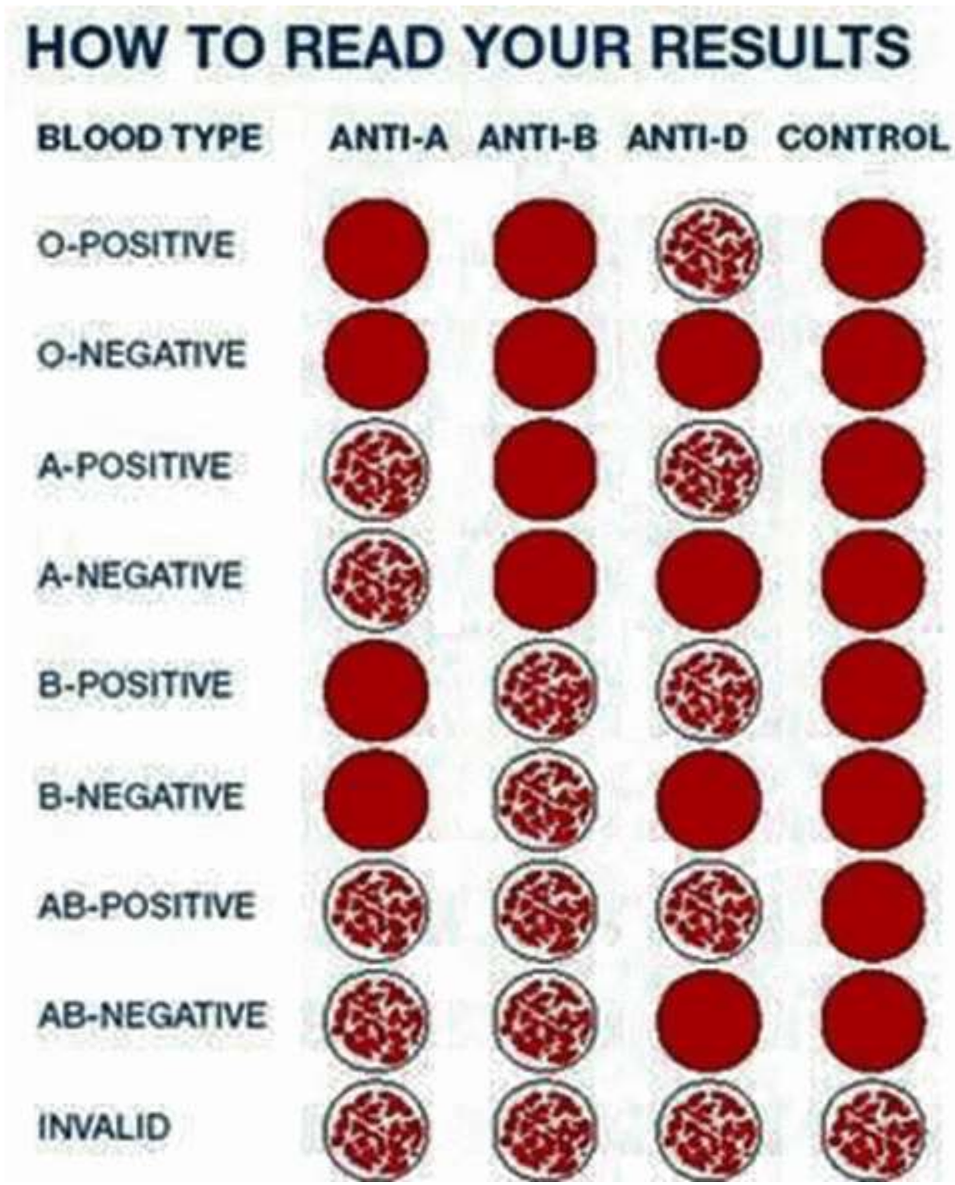


Fig.8.5: Procedure of Blood typing(grouping)

Possible Results



CUATIONS

- Use the blood lancet one time.
- Don’t touch the tapered end of the blood lancet.
- Do this experiment under the instructor supervision.
- Dispose the waste of this experiment the instructor guide you.

THE END